Venkata Lakshmi Parimala Pasupuleti

AI Safety & Guardrails · LLM Evaluation · Fair Housing Compliance · AgentProbe · Prompt Injection Defense · Python · Terraform · AWS

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PROFESSIONAL SUMMARY

AI Safety & Evaluation Engineer who brings two independent proof points that are rarely found in the same candidate. (1) **Eval infrastructure:** built AgentProbe - a homegrown LLM evaluation toolkit with LLM-as-judge classifiers, 5-metric engine, CI-integrated regression harness, and golden datasets - the same architecture RealAssist's guardrail pipeline requires. (2) **Fair Housing domain:** spent 7 years owning HUD compliance pipelines at Entrata (income limits, eligibility, HAP, Tax Credit calculations) - the same Fair Housing Act obligations governing every RealAssist response. Also brings: prompt injection defense, OWASP LLM Top 10, red-teaming, content moderation, precision/recall/F1 evaluation across 1,240+ labeled test cases, and Terraform/IaC. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and data pipelines; integrating AI safety components across backend and reporting layers; debugging cross-stack reliability issues in data-driven features.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern · Remote

- Engineered validation guardrails enforcing FAA altitude limits, battery budgets, and geometric no-fly-zone checks - blocking non-compliant missions before export; every compliance check functions as a hypothesis test grounded in source regulation.
- Built RAG layer over FAA Part 107 regulatory corpus with cited-source answers; implemented LangGraph stateful pipeline with conditional routing (compliant → export, non-compliant → rejection with grounded reasons).

George Mason University
Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); hypothesis testing, confidence intervals, and model calibration. HuggingFace Transformers fine-tuning + inference API deployment; anomaly detection (Isolation Forest). Workshops to 600+ students.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Owned HUD Fair Housing compliance pipelines for 7 years** - annual income-limit import (few thousand federal rows → ~180,000 domain records) powering eligibility, rent, HAP, and Tax Credit calculations for 100K+ housing units. Every release included baseline validation and reconciliation - same Fair Housing Act obligations governing RealAssist. Zero compliance violations across the entire ownership period.
- **Built and maintained PHI-compliant, audit-ready data pipelines** - Libsodium-encrypted AWS S3 document migration (100% PHI compliance, zero security incidents); RabbitMQ async processing with idempotency guarantees; audit logging across all resident-data workflows. Owned 20+ production modules (Python, PHP, PostgreSQL, AWS) at 99.5% uptime across 120+ Agile sprints.
- **Instrumented production systems for observability and safety** - OpenTelemetry + Prometheus + Grafana (40% reduction in debugging time); 200+ SQL optimizations (30% performance gain); Python/Bash automation eliminating 80% of manual operational workload; mentored 5+ engineers; 300+ code reviews.

Entrata India, Pune
Jul 2017 – Sep 2018

Associate Software Engineer

- Built affordable housing compliance modules (PHP, Python, Java, PostgreSQL); income limits import (1.6M HUD/Tax Credit records); AWS S3 migration; document management and certification workflows.

PROJECTS

AgentProbe: LLM Evaluation & Guardrail Infrastructure
LLM-AS-JUDGE · CI-INTEGRATED · OPEN SOURCE

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- Homegrown AI evaluation toolkit: 15-endpoint FastAPI backend; 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate); **LLM-as-judge classifiers** with configurable scoring rubrics; **CI-integrated regression harness** with golden datasets and GitHub Actions pass/fail gating - guardrails can't silently degrade; offline + online evaluation loops; distributed model routing across OpenAI/Anthropic/Groq/Ollama. Precision/recall/F1, confusion matrix analysis, inter-rater agreement, and dataset synthesis built in.

Leasel: Fair Housing Compliance & Guardrailed AI Platform
FAIR HOUSING ACT · 93.6% PRECISION

Python · LangGraph · FastAPI · TF-IDF · OCR · SQLite · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- 8-agent LangGraph platform enforcing Fair Housing Act guardrails on AI-generated responses - the same regulatory domain as RealAssist. Core safety rule enforced in code: **LLM extracts and reasons; a human or deterministic logic makes every regulated compliance call** - tenant eligibility questions always route to staff, never resolved by AI. Achieved **93.6% precision and 87.9% F1** across 10 Fair Housing compliance rules on 1,240 residential lease PDFs (5 US states + federal); pass@k harness with multi-trial sampling, LLM-as-judge scoring, and regression gate. Missing/conflicting clause detection; human-review gate on all regulated decisions; citation grounding to exact source clauses.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

AI Safety & Evaluation: LLM-as-Judge Classifiers, Guardrails, Content Moderation, Prompt Injection Defense, OWASP LLM Top 10, Jailbreak Defense, Red-Teaming, Input/Output Filtering, Eval-Driven Development, Precision/Recall/F1, Confusion Matrix, Golden Datasets, CI Regression Harnesses, Hallucination Detection, Dataset Synthesis, Audit Logging, PHI Compliance, HIPAA, SOC 2

LLM & Agents: Python (8+ yrs), LangGraph, LangChain, RAG Pipelines, LLM Evaluation, AgentProbe, Agentic Workflows, FastAPI, OpenAI API, Anthropic API, Groq, Ollama, LangSmith, Prompt Engineering, Few-Shot Prompting, Structured Outputs, Semantic Search, Embeddings, ChromaDB

Cloud, IaC & Observability: AWS (S3, EC2, RDS, Lambda, Step Functions, Bedrock, SageMaker), Terraform, Docker, Kubernetes, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana, PostgreSQL, Redis, Apache Kafka, Google Cloud Platform (GCP), Vertex AI